

Java Priority Queue

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In computer science, a priority queue is an abstract data type which is like a regular queue, but where additionally each element has a "priority" associated with it. In a priority queue, an element with high priority is served before an element with low priority. - Wikipedia

In this problem we will test your knowledge on Java Priority Queue.

There are a number of students in a school who wait to be served. Two types of events, ENTER and SERVED, can take place which are described below.

- ENTER: A student with some priority enters the queue to be served.
- SERVED: The student with the highest priority is served (removed) from the queue.

A unique id is assigned to each student entering the queue. The queue serves the students based on the following criteria (priority criteria):

1. The student having the highest Cumulative Grade Point Average (CGPA) is served first.
2. Any students having the same CGPA will be served by name in ascending case-sensitive alphabetical order.
3. Any students having the same CGPA and name will be served in ascending order of the id.

Create the following two classes:

- The Student class should implement:
 - The constructor `Student(int id, String name, double cgpa)`.
 - The method `int getID()` to return the id of the student.
 - The method `String getName()` to return the name of the student.
 - The method `double getCGPA()` to return the CGPA of the student.
- The Priorities class should implement the method `List<Student> getStudents(List<String> events)` to process all the given events and return all the students yet to be served in the priority order.

Input Format

The first line contains an integer, `n`, describing the total number of events. Each of the `n` subsequent lines will be of the following two forms:

- ENTER name CGPA id: The student to be inserted into the priority queue.
- SERVED: The highest priority student in the queue was served.

The locked stub code in the editor reads the input and tests the correctness of the Student and Priorities classes implementation.

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Difficulty: Medium
Max Score: 20
Submitted By: 71679

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```
1 import java.util.ArrayList;
2 import java.util.List;
3 import java.util.Scanner;
4 import java.util.PriorityQueue;
5
6 class Student {
7     private int id;
8     private String name;
9     private double cgpa;
10
11     public Student(int id, String name, double cgpa) {
12         this.id = id;
13         this.name = name;
14         this.cgpa = cgpa;
15     }
16
17     public int getID() {
18         return id;
19     }
20
21     public String getName() {
22         return name;
23     }
24
25     public double getCGPA() {
26         return cgpa;
27     }
28 }
29
30 class Priorities {
31
32     public List<Student> getStudents(List<String> events) {
33
34         PriorityQueue<Student> queue = new PriorityQueue<>(
35             (a, b) -> {
36                 // Higher CGPA first
37                 int result = Double.compare(b.getCGPA(), a.getCGPA());
38             }
```

Line: 1 Col: 28

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Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

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```
1 12
2 ENTER John 3.75 50
3 ENTER Mark 3.8 24
4 ENTER Shafaet 3.7 35
5 SERVED
6 SERVED
7 ENTER Samiha 3.85 36
8 SERVED
9 ENTER Ashley 3.9 42
10 ENTER Maria 3.6 46
11 ENTER Anik 3.95 49
```